清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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Assessment of Iron and Steel Technologies from A Global Perspective

Tsinghua University & Saudi Aramco

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清华大学气候变化与可持续发展研究院
Institute of Climate Change and Sustainable Development,Tsinghua University

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January 12, 2026

Project Timeline

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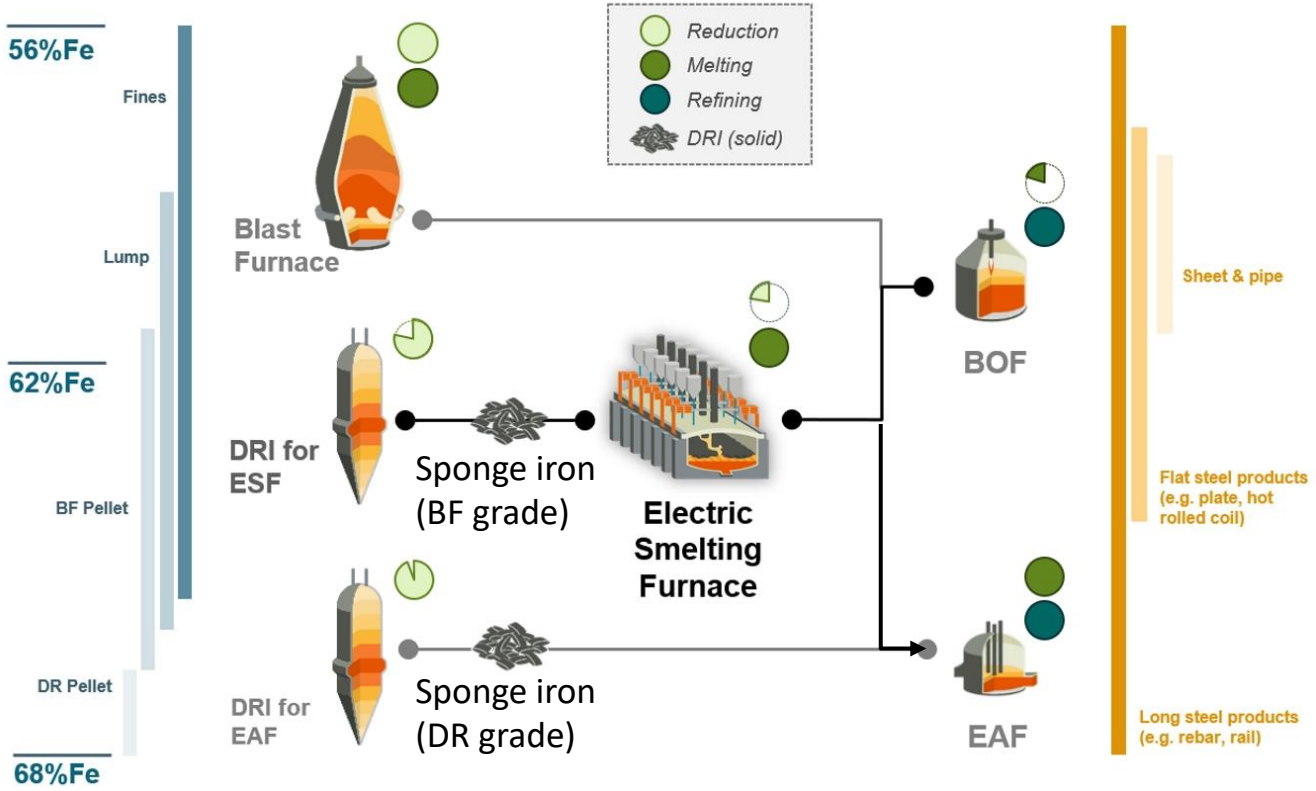
	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July	Aug	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	June	July
Research Proposal & preparations																								
Overview of global steel trade & technology assessment																								
Data requirement & database investigation (Base on case India)																								
Model upgrade (Base on case India)																								
Case study: India																								
Case study: USA & Saudi Arabia																								
Case study: Australia & Japan																								
Case study: Germany & Russia																								
Case study: China-Australia steel trade route																								
Final report																								

Research Update

- Supplementary information: H2-DRI-ESF-BOF/EAF route in India
- Model refinement and presentation
- Update results for India

Review: H2-DRI-ESF-EAF/BOF route

- Traditional H2-DRI-EAF rely on high iron content (>67.5%) and low gangue content (<1.5%).
- By incorporating an **ESF furnace** between DRI and EAF for **supplementary melting** and **slag removal**, the H2-DRI-ESF-BOF/EAF route can support **lower-grade iron ores**.
- If the goal is to achieve decarbonization while reducing the reliance on ultra-high-grade iron ore, the H2-DRI-ESF-BOF/EAF process is a technology path worth considering.



Tech route	Ore Fe%	Ore Slag%
H2-DRI-EAF	**≥67-68%**	**<2-3%**
Gas-DRI-EAF	≥67%	<2-3%
Coal-DRI-EAF	≥64-65%	<4-6%
H2-DRI-ESF-BOF/EAF	**≥62-65%**	**/**
BF-BOF	≥58%	/

Source: Thyssenkrupp, BHP Billiton

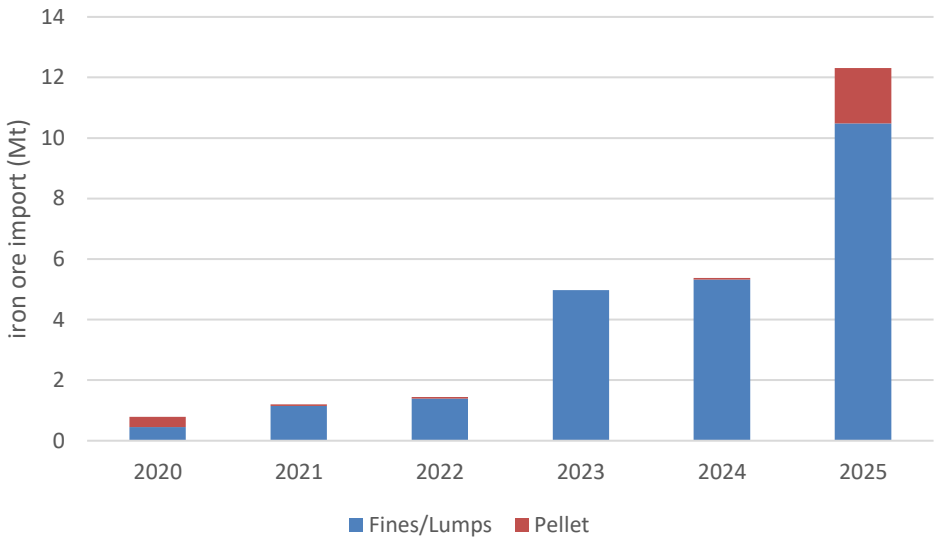
H2-DRI-ESF-EAF/BOF route in India

- India's own ore resources are insufficient to support the production of high-grade pellets suitable for **H2-DRI**, and can only be used for **Coal-DRI**.
- In the future, India is likely to **rely on imports** of high-grade ore from Brazil and Oman.
- **Developing the H2-DRI-ESF-EAF/BOF route rather than H2-DRI-EAF in India is a considerable option.**

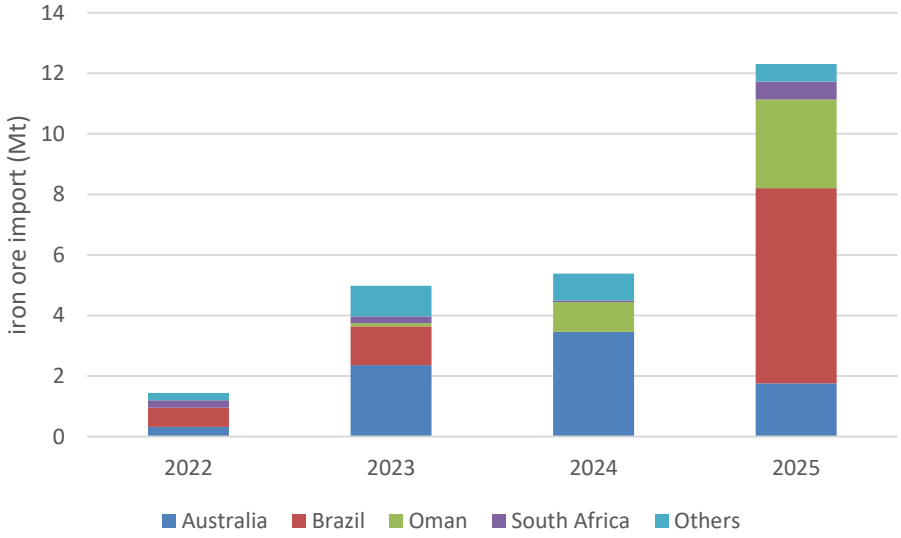
India's domestic iron ore supplement

Types of ore	Reserves	Pros	Cons
Hematite	7004Mt	High **Fe% content** of ~60-67%	High **slag content** and **Al_2O_3 rate**, hard for beneficiation
Magnetite	58.50Mt	Great resources totaling **10561Mt**	Low reserves, Shortage of **Metallurgical-grade ore(0.4Mt)**

India's iron ore imports: by type

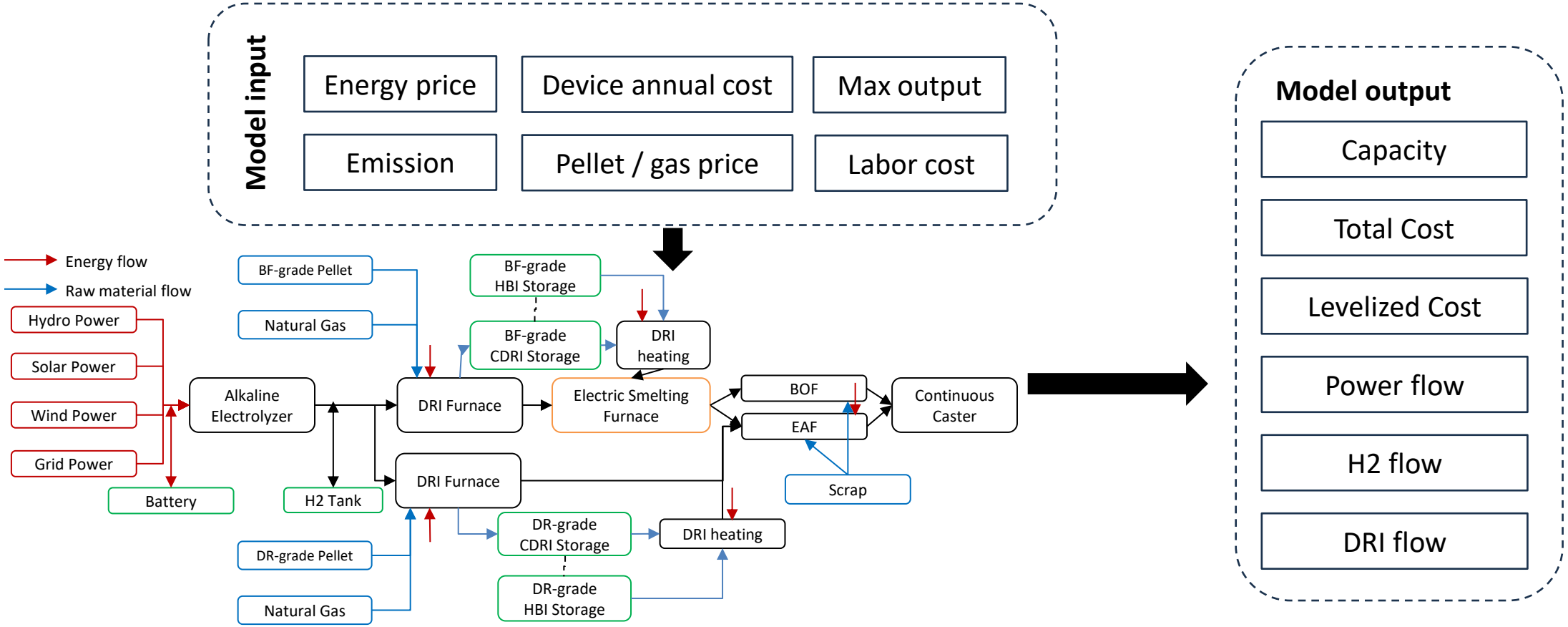


India's iron ore imports: by country



Methods: Current model

- The new model establishes **relatively independent pathways** for both BF-grade and DR-grade materials.
- By applying appropriate constraints, the model can simulate the competition and synergy among multiple routes: **BF-grade DRI-ESF-BOF**, **BF-grade DRI-ESF-EAF**, and **DR-grade DRI-EAF**.
- Programming language: **Switching GAMS->Python**; Optimization solver: **COPT**



Methods: Current model

Model: Input parameters / dataset

Category	Item / Tech	Parameter	Resolution	Unit
Energy	Solar	Max output	Hourly	/
		Annual cost	Annual	M USD/MW
	Wind	Max output	Hourly	/
		Annual cost	Annual	M USD/MW
	Grid	price	Annual	M USD/MWh
		emission	Annual	MtCO2/MWh
	Hydro	price	Hourly	M USD/MWh
		usage rate	Hourly	/
Feedstock	Iron ore	**BF pellet price**	**Annual**	**USD/t**
		DR pellet price	**Annual**	**USD/t**
	Scrap	Scrap price	Annual	USD/t
	Gas	Gas price	Annual	M USD/MWh
		Gas emission	Annual	MtCO2/MWh

Category	Item / Tech	Parameter	Resolution	Unit
Device	Electrolyzer	Annual cost	Annual	M USD/MW
	H2 Tank	Annual cost	Annual	M USD/MW
	Battery Power	Annual cost	Annual	M USD/MW
	Battery Energy	Annual cost	Annual	M USD/MW
	DRI Furnace	Annual cost	Annual	MUSD/ t/h
	EAF Furnace	Annual cost	Annual	MUSD/ t/h
	ESF Furnace	Annual cost	Annual	MUSD/ t/h
	BOF Furnace	Annual cost	Annual	MUSD/ t/h
	HBI Storage	Annual cost	Annual	MUSD/t
	CDRI Storage	Annual cost	Annual	MUSD/t
	Labor	Ironmaking cost	Annual	MUSD/t
		Steelmaking cost	Annual	MUSD/t

- Most of this data uses local Indian or South Asian data, with a small portion coming from public databases, proprietary databases, and relevant research reports.
- Some data, which we believe can use global default values (such as equipment power consumption), is not included in this table.

Methods: Current model

Model: Output value

Category	Output	Resolution	Unit
Capacity	wind	Annual	MW
	solar	Annual	MW
	Battery (power)	Annual	MW
	Battery (energy)	Annual	MW
	Electrolyzer	Annual	MW
	Tank	Annual	kg
	DRI furnace	Annual	t
	EAF furnace	Annual	t
	ESF furnace	Annual	t
	BOF furnace	Annual	t
Levelized cost	LCOE (Electricity)	Annual	USD/MWh
	LCOH (Hydrogen)	Annual	USD/kg
	LCOD (DRI)	Annual	USD/t
	LCOS (Steel)	**Annual**	**USD/t**

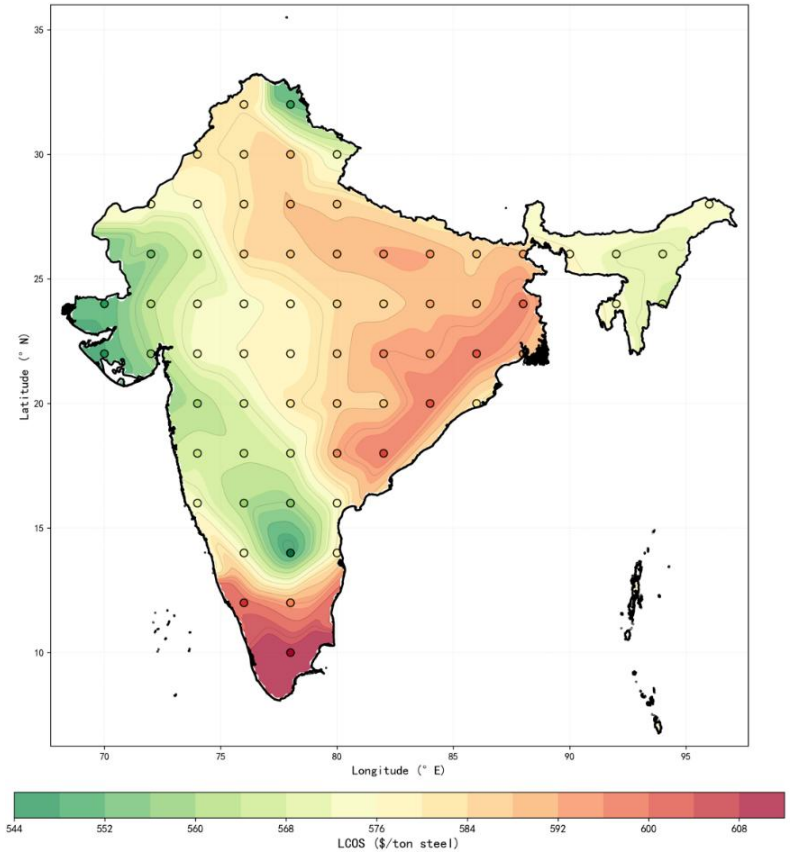
Category	Output	Resolution	Unit
Total Cost	Wind	Annual	M USD
	Solar	Annual	M USD
	Battery	Annual	M USD
	Electrolyzer	Annual	M USD
	H2 Tank	Annual	M USD
	DRI furnace	Annual	M USD
	EAF furnace	Annual	M USD
	Iron ore (BF/DR)	Annual	M USD
	Scrap cost	Annual	M USD
	Gas cost	Annual	M USD
	Gird power cost	Annual	M USD
	Hydropower cost	Annual	M USD
	Other material cost	Annual	M USD
	IronLabor cost	Annual	M USD
	SteelLabor cost	Annual	M USD
	Total cost	Annual	M USD

Category	Output	Resolution	Unit
Power flow	Solar	Hourly	MW
	Wind	Hourly	MW
	Grid	Hourly	MW
	Hydro	Hourly	MW
	Electrolyzer usage	Hourly	MW
	DRI usage	Hourly	MW
	ESF usage	Hourly	MW
	EAF usage	Hourly	MW
	BOF usage	Hourly	MW
	Reheat usage	Hourly	MW
	Battery discharge	Hourly	MW
	Battery charge	Hourly	MW
	Battery storage	Hourly	MWh
H2 flow	Electrolyzer production	Hourly	kg
	H2 Tank charge	Hourly	kg
	H2 Tank discharge	Hourly	kg
	DRI usage	Hourly	kg
	H2 Tank storage	Hourly	kg
DRI flow	DRI production(BF/DR)	Hourly	t
	DRI from H2(BF/DR)	Hourly	t
	DRI from gas(BF/DR)	Hourly	t
	DRI charge(BF/DR)	Hourly	t
	DRI discharge(BF/DR)	Hourly	t
	EAF/ESF DRI(BF/DR)	Hourly	t
	DRI storage(BF/DR)	Hourly	t
	DRI ironore(BF/DR)	Hourly	t

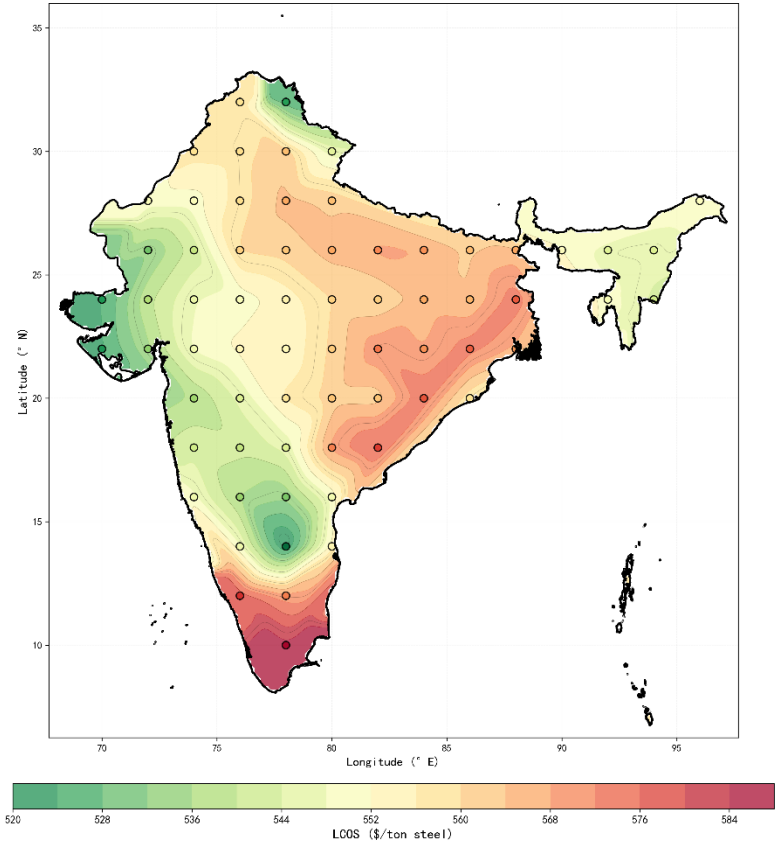
- Generally, we are most concerned with the level of LCOS (Levelized Cost of Storage) and the key factors that influence it.

Preliminary analysis: BF & DR

- LCOS decreased for about **25 USD.**
- Since the power output curve and load level did not change, the distribution of the results did not change so significantly.



LCOS heatmap of the H_2-DRI-EAF route in major regions of India (USD/tcs)
Using DR-pellet with 168USD/t



LCOS heatmap of the H_2-DRI-EAF route in major regions of India (USD/tcs)
Using BF-pellet with 93USD/t

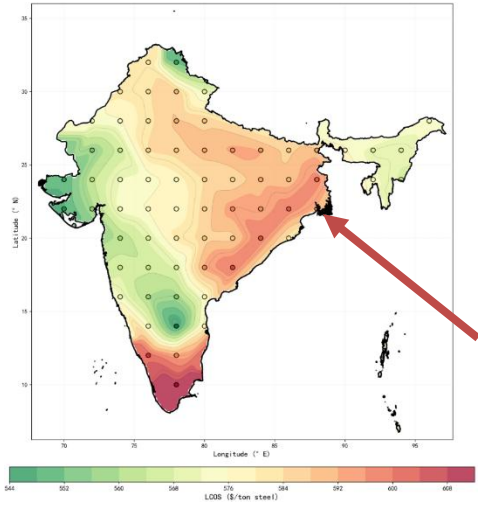
India's iron ore supply prices

Type	price	Source
import DR pellet	$158-168	Estimates based on the Platts pellet premium
import BF pellet	$112-118	CFR Kandla / Mundra
domestic BF pellet	$88-98	Odisha Index

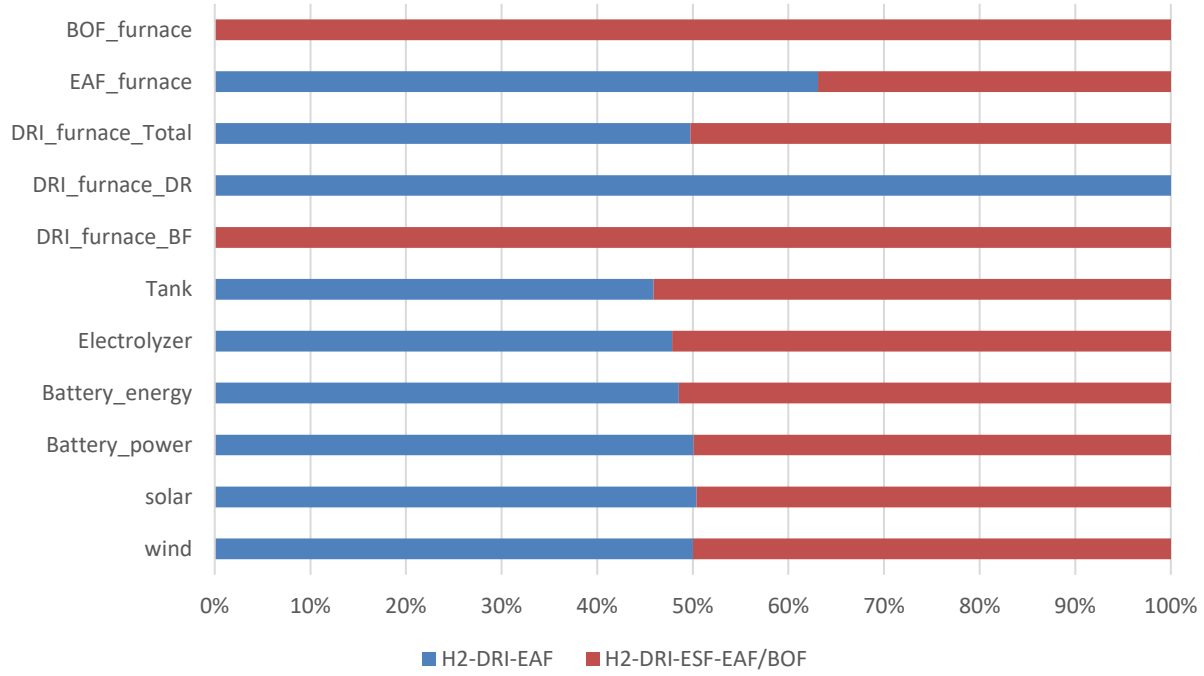
Preliminary analysis: H2-DRI-EAF vs H2-DRI-ESF-...

Comparative analysis

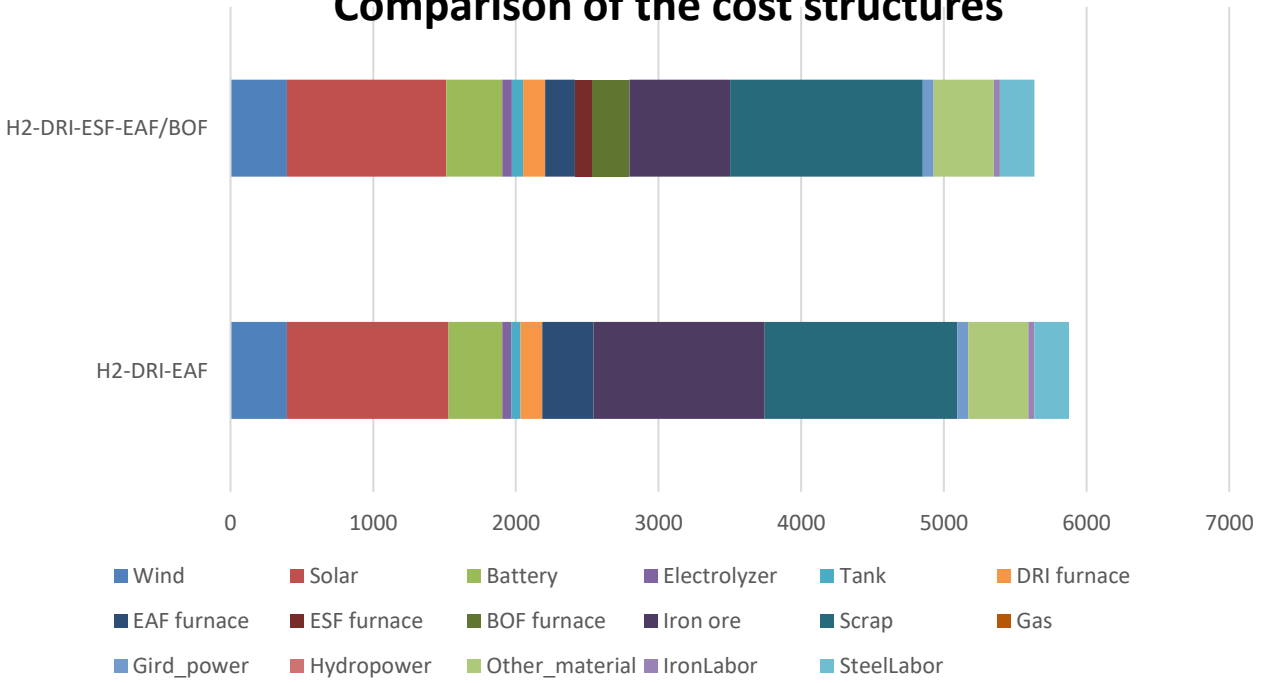
- The **power consumption** of the H2-DRI-ESF-EAF/BOF process did not show a significant increase, which may be due to not considering the **additional energy consumed for slag formation** and not accounting for the small amount of **carbon added in the ESF**.
- The H2-DRI-ESF-EAF/BOF process **utilizes both BOF and EAF** instead of one completely replacing the other, which may be due to the participation of **lower-priced scrap**.



The relative proportion of capacity



Comparison of the cost structures



Next steps

- Perform uncertainty analysis on the prices of scrap, BF-pellet, and DR-pellet.
- Evaluate the impact of the carbon added to the ESF on the overall process.
- Simulating the brownfield redevelopment scenarios of existing BOF/EAF facilities.
- Explore the relationship between BF and DR prices, when the total costs of the two routes are approximately equal.
- Identifying the specific locations of steel production plants, mines, and other facilities in India, and then Exploring methods for simulating the spatial distribution of iron ore prices.
- Case study: Long-Term Transformation of India's Steel Industry

Thank You For Your Attention!